

Baseline Labs Summary

PathCare results — 1 December 2022

To: Dr. Aleatia Burger (GP, women's health)
From: Elvera van Rooyen
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Companion to: Pre-consultation Summary — Perimenopause / HRT workup

Note on preparation: This summary was prepared by the patient in collaboration with a Claude-based AI clinical research assistant. The original PathCare reports (Haematology, Endocrinology, Biochemistry) are attached separately as the source documents. This summary is offered to assist interpretation, not to replace clinical judgement.

1. Context

These labs were ordered by Dr A van der Velde (holistic GP, Cape Town) on 1 December 2022. I was 41 at the time, on copper IUD (no exogenous hormones), and already on Epitec (lamotrigine) and Rivotril (clonazepam) for TLE / dystonia. I was not perimenopausal at this point — symptoms only began ~September 2025. These therefore represent a useful **pre-perimenopausal baseline** with the added advantage that the timing of the blood draw was favourable for hormone interpretation:

- **LMP:** 8 November 2022
- **Blood draw:** 1 December 2022
- **Cycle day:** ~day 23 — mid-to-late luteal phase, where progesterone should peak. This means the progesterone reading is genuinely informative, not a random snapshot.

2. Headline clinical findings

The fuller picture is in section 3. In summary:

- **Robust ovulatory cycle.** Late-luteal progesterone 57.9 nmol/L on day ~23. Estradiol 733 pmol/L. The corpus luteum was working well. This is important: it shows that what I'm experiencing now (mood, sleep, irritability, weight pattern) represents a real **decline** from documented prior function, not a baseline weakness.
- **Excellent metabolic profile.** Fasting insulin 2.2 mIU/L (flagged Low — this is good news, not a problem; it reflects keto-induced insulin sensitivity). QUICKI 0.444 (excellent). Fasting glucose 4.5 mmol/L. Lipids and ApoB were not ordered in 2022.
- **Very low inflammation.** hs-CRP <0.4 mg/L (low-risk band).
- **Euthyroid.** TSH 1.45 mIU/L (near optimal), free T4 12.9, free T3 3.5 (low end of range). Antibodies and reverse T3 not done.
- **Good iron and B12 stores.** Ferritin 103 µg/L, B12 479 pmol/L.
- **Two soft signals worth attention** (detailed below): vitamin D 26 ng/mL (just into "optimal" band, functionally low for this patient profile) and homocysteine 10.4 µmol/L (upper-normal — lamotrigine is a recognised cause).
- **Low Free Androgen Index (1.0%)** — bioavailable testosterone sitting at the bottom of the normal range even in 2022.

3. Detailed breakdown by domain

3.1 Reproductive hormones (day ~23 of cycle)

Test	Result	Reference	Comment
Estradiol (E2)	733 pmol/L	Luteal 140–2382	Solidly mid-luteal. Endogenous estrogen production was robust.
Progesterone	57.9 nmol/L	Luteal > ~16 confirms ovulation	Robust ovulatory progesterone. Almost 4× the ovulation cut-off. Corpus luteum functioning well.
LH	3.04 IU/L	Luteal 1–14	Normal premenopausal pattern.
FSH	Not measured	—	Gap. Essential to measure now as the key peri marker.
Total testosterone	1.1 nmol/L	0.5–1.9	Normal.
SHBG	107 nmol/L	12–137	High-normal. Explains the low free androgen index below.
Free Androgen Index	1.0%	0.7–8.7%	At the very bottom of normal. Bioavailable testosterone was already low in 2022. Worth tracking; testosterone replacement may eventually be worth considering as a third HRT pillar.
DHEA-S	8.0 µmol/L	2.0–11.1	Normal mid-range adrenal androgen.
AMH	Not measured	—	Gap. Useful in current peri staging.

3.2 Thyroid

Test	Result	Reference	Comment
TSH	1.45 mIU/L	0.35–4.94	Near optimal. Functionally optimal is 1.0–2.0; this sits in that window.
Free T4	12.9 pmol/L	9.0–19.1	Normal mid-range.
Free T3	3.5 pmol/L	2.4–6.0	Low end of normal — flagged for awareness; worth tracking with reverse T3 now.
Reverse T3	Not measured	—	Gap. Would tell us if conversion is being suppressed.
Anti-TPO	Not measured	—	Gap. Important to rule out autoimmune thyroiditis, especially in peri.
Anti-Thyroglobulin	Not measured	—	Gap, same reason.

3.3 Metabolic / insulin sensitivity

Test	Result	Reference	Comment
Fasting glucose	4.5 mmol/L	3.9–5.5	Normal.
Fasting insulin	2.2 mIU/L (flagged L)	3.0–17.0	Excellent. The "Low" flag reflects a reference range built for an insulin-resistant general population. 2.2 indicates high insulin sensitivity from sustained keto.
QUICKI Index	0.444	>0.357 = good	Excellent insulin sensitivity.

Test	Result	Reference	Comment
HbA1c	Not measured	—	Gap. Longer-term glucose picture worth adding.
ApoB	Not measured	—	Gap. The right marker for cardiovascular tracking on keto.
Full lipid panel	Not measured	—	Gap. Important baseline before HRT.

3.4 Inflammation, liver, kidney

Test	Result	Reference	Comment
hs-CRP	<0.4 mg/L	<1 = low risk	Very low inflammation.
GGT	23 IU/L	<34	Normal.
ALT	30 IU/L	<36	Normal upper. Liver tolerating lamotrigine well.
AST	24 IU/L	<36	Normal.
Creatinine	73 µmol/L	50–98	Normal.
eGFR (CKD-EPI)	88 mL/min/1.73m ²	≥90	Normal kidney function.
Urea	3.2 mmol/L	2.1–7.1	Normal.
Sodium / Potassium / Chloride / Bicarbonate / Anion gap	138 / 4.2 / 101 / 24.8 / 12	All within range	Electrolytes balanced.

3.5 Micronutrients & methylation

Test	Result	Reference	Comment
Ferritin	103 µg/L	30–200	Healthy iron stores despite heavier bleeding on copper IUD.
Vitamin B12	479 pmol/L	>250 unlikely deficient	Comfortable.
Vitamin D (25-OH)	26 ng/mL	Optimal >20 (lab); functionally ≥40	Just into the lab's "optimal" band but on the low side functionally. Most evidence-current targets for bone, immune, mood, and hormone outcomes in peri are 40–60 ng/mL. Worth raising via D3 + K2 supplementation; remeasure.
Homocysteine	10.4 µmol/L	4.4–13.6	Upper-normal. The PathCare report explicitly lists anticonvulsants as a cause of moderate elevation — lamotrigine is the likely driver via methylation/folate pathway interference. Estrogen lowers homocysteine; HRT would be expected to reduce it. Methylated B-complex (methylfolate, methyl-B12, P5P) worth considering.
PTH	2.6 pmol/L	1.6–6.9	Normal. Calcium homeostasis intact.
Magnesium RBC, zinc, selenium	Not measured	—	Gaps. Useful on carnivore and in peri.

3.6 Haematology (Full Blood Count)

Entirely normal. Hb 14.0 g/dL, Hct 0.41, MCV 93, MCH 32, MCHC 34, RDW 12.3, WCC 6.3, neutrophils 67.5%, lymphocytes 23.5%, platelets 306. No flags. Not anaemic, no morphological abnormalities, normal immune cell distribution.

3.7 Adrenal

Test	Result	Reference	Comment
Cortisol (random)	373 nmol/L	10am: 102–535	Normal random cortisol. Important context: this was drawn during the early phase of the 2022–Aug 2025 high-stress period (self-rated 9/10), so I would not have expected an obviously suppressed pattern at this point.

4. What this means for the current workup

The most clinically useful single point: robust luteal-phase progesterone of 57.9 nmol/L in 2022 gives us an anchor to compare a new day-21 progesterone against. If the new value is materially lower (or anovulatory), that is objective biochemical confirmation of the perimenopausal transition and of the specific deficit (allopregnanolone-mediated GABAergic destabilisation) driving much of the current symptom picture.

What I'd value comparing in the new labs:

- Day-21 progesterone vs 57.9 nmol/L (2022).
- Estradiol vs 733 pmol/L (luteal, 2022) — with the caveat that timing matters.
- FSH (new) and AMH (new) for ovarian reserve and peri staging.
- TSH and the full thyroid panel vs 2022 (TSH 1.45, fT4 12.9, fT3 3.5), plus the antibodies that weren't done.
- Fasting insulin, QUICKI vs 2.2 / 0.444 — to confirm the metabolic flexibility hasn't shifted.
- hs-CRP vs <0.4 — to confirm inflammation is still well-controlled.
- Ferritin vs 103 µg/L — given the recent bleeding pattern change, this needs re-checking.
- Homocysteine vs 10.4 µmol/L — track methylation status and the effect of any B-vitamin supplementation.
- Vitamin D vs 26 ng/mL — ideally now ≥40.
- Free Androgen Index vs 1.0% — if testosterone discussion becomes relevant.

Gaps in the 2022 panel that should be filled this time:

- FSH and AMH (peri-specific markers).
- Full lipid panel and ApoB (cardiovascular completeness on keto).
- HbA1c (long-window glucose).
- Reverse T3, anti-TPO, anti-thyroglobulin (full thyroid).
- Magnesium RBC, zinc, selenium (micronutrient completeness).
- Baseline lamotrigine serum level (pre-HRT reference value).

5. One-paragraph summary

In December 2022 (age 41, pre-peri, on copper IUD), my labs showed a functioning ovulatory cycle with robust luteal progesterone (57.9 nmol/L) and healthy estradiol (733 pmol/L), excellent insulin sensitivity

(QUICKI 0.444, fasting insulin 2.2 mIU/L), very low inflammation (hs-CRP <0.4), normal thyroid (TSH 1.45), good iron and B12 stores, and unremarkable liver, kidney and haematology. Soft signals worth noting were a borderline vitamin D (26 ng/mL), an upper-normal homocysteine (10.4 μ mol/L — most likely lamotrigine-related), and a Free Androgen Index at the bottom of normal (1.0%). Several useful tests were not ordered (FSH, AMH, ApoB, full lipid panel, HbA1c, reverse T3, anti-TPO, anti-Tg, magnesium RBC) — these are now in the proposed panel for the current workup.

Source documents: PathCare Haematology, Endocrinology, and Biochemistry reports, all dated 1 December 2022 (Specimen IDs 1201:HA01814R, 1201:EA00844R, 1201:BA02338R, Requisition 656907686). Attached separately.